

Professional Summary

Computational physicist with a PhD in Physics and ten years of experience building and running simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led an upgrade of the PatchworkMHD code on Frontera and used gradient-based optimization to calibrate model parameters against observations. Nine publications, three as first author, and Co-I on a funded NASA Fermi grant.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, Lagrangian particle dynamics, numerical PDEs (advection-diffusion, Fokker-Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Geospatial & Remote Sensing: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

Languages: Python, C/C++, Fortran 95, R, SQL, Shell, Rust, Java, Julia

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Education

Ph.D. in Physics

Universitat de València, Spain

ADVISOR: Prof. Miguel A. Aloy & Dr. Petar Mimica.

THESIS: *Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows*

Oct. 2011 – Jul. 2017

Excellent *cum laude*.

M.Sc. in Physics

Universidad Michoacana de San Nicolás de Hidalgo, Mexico

ADVISOR: Prof. José A. Cervera

THESIS: *Study of TOV stars with the SPH method*

Aug. 2009 – Sep. 2011

B.Sc. in Physics

Universidad Autónoma del Estado de México, Mexico

ADVISOR: Prof. Francisco S. Guzmán

THESIS: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Aug. 2004 – Dec. 2008

“Dr. Juan Josafat Pichardo” Award

Experience

Independent Contractor

Mercor, Seattle, WA, USA

Jan 2026 — Present

- Provide expert-level knowledge in Physics, Astrophysics, and Mathematics to improve models for a top AI research lab
- Assess model reasoning quality, identify failure modes, and provide structured feedback to improve scientific accuracy
- Apply understanding of mathematical foundations to benchmark model performance, and quality assurance on complex STEM problems

Spatial Data Scientist

TealWaters, Seattle, WA, USA

May 2025 — Jan 2026

- Lead project to optimize code for topographic terrain modeling and elevation derivative calculation.
- Engineered Python tools for geospatial data processing and analysis, utilizing QGIS for visualization and analysis.
- Utilize Wetland Intrinsic Potential (WIP) tool with random forest for wetland probability mapping in Skykomish watershed.
- Performed EDA, feature engineering, and geospatial data preparation, including multispectral (Sentinel 1/2) processing for ML land-cover and wetland probability models.
- Share knowledge and results with managers and decision-makers, train team members on WIP tool usage, and collaborate with software engineers and scientific team to transition prototypes into production.

Software Engineer

Apr 2022 – Jan 2024

Paychex, Rochester NY, USA; based in Seattle, WA, USA

- Developed Java Kafka consumers for high-throughput data transfer across enterprise databases, and deployed to production via Jenkins and OpenShift using CI/CD pipelines.
- Developed Python tests and CI pipelines ensuring software quality standards.
- Conducted SQL data preparation, validation, and analysis from Oracle EBS datasets.

Postdoctoral Research Associate

Feb 2021 – Apr 2022

Rochester Institute of Technology, Rochester, NY, USA

- Led a team of specialists on a NSF-sponsored project to upgrade the C code PatchworkMHD to perform HPC simulations using state-of-the-art numerical techniques to model supermassive black hole binaries.
- Implemented a new feature (black hole spin) to PatchworkMHD, making more realistic binary black hole simulations without impacting runtime.
- Designed the experiments and evaluated state-of-the-art mathematic and numerical algorithms implemented in PatchworkMHD by running simulations at Frontera supercomputer (TACC, UT at Austin).
- Worked in a detail-oriented manner to successfully benchmark and identify performance optimization opportunities of the scientific code.
- Calibrated Paramo model parameters against blazar spectral energy distributions observed by the Fermi-LAT telescope with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem, demonstrating ML-assisted parameter estimation workflows for complex astrophysical systems.
- Participated in a multi-institutional collaboration to study binary Neutron Star mergers through HPC simulations, resulting in 2 publications that provided critical breakthrough insights of the physics underlying these events.
- Published 3 co-authored papers and mentored graduate students (2 Ph.D.).

Postdoctoral Research Fellow

Oct 2018 – Nov 2020

Purdue University, West Lafayette, IN, USA

- Developed the open-source code, Paramo, a numerical code in Fortran 95 optimized with OpenMP to perform radiative transfer simulations in relativistic astrophysics scenarios.
- Served as a Co-I on two NASA Fermi Guest Investigator Program proposals, one granted. The awarded grant helped with research on our understanding of the two main types of blazars, identifying that important physical constraints applied to both objects.
- Developed Python tools to calculate the loss of energy due of high-energy particles due to interactions spectrum and evolution in the context of gamma-ray burst afterglows by developing sophisticated numerical integration, and OpenMP optimized features to Paramo.
- Collaborated with a group of multidisciplinary scientists to develop Python scripts for statistically modeling the COVID-19 outbreak in Mexico and helped create scientific infographics and blogposts for Spanish-speaking populations to reduce the spread of misinformation.
- Published 1 first-author and 1 co-authored paper and mentored three graduate students (1 M.S. and 2 Ph.D.).

Postdoctoral Research Fellow

Jan – Sep 2018

Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacan, Mexico

- Contributed to a Python pipeline to process 2D simulation images from the GRTrans numerical code, converting outputs to grayscale, normalizing pixel intensities, and extracting spatial features to generate training data for an SVM classifier predicting black hole spin from radio image morphology.
- Developed an open-source data analysis and visualization tool in Python to provide any user with accessible tools to calculate radiative transfer phenomena (spectra and light-curves) in relativistic astrophysics.
- Organized a workshop to train graduate students in the use of the high-volume data storage tool HDF5.

Graduate Research Assistant

Oct 2011 – Jul 2017

Universitat de València, Burjassot, Valencia, Spain

- Independently developed Shell and Python scripts to build pipelines to run simulation of the radiative transfer code C-SPEV and perform data processing of output datasets in HDF5 format, ensuring data quality and integrity for downstream analysis and model fit with observations.
- Developed Python scripts to perform exploratory data analysis on multi-wavelength observational datasets from the NASA Fermi-LAT telescope and the Very Large Baseline Array (VLBA), including sky-coordinate-based source cross-matching across catalogs and log-parabola spectral fitting to characterize blazar emission.
- Independently constructed models from C-SPEV simulations for curve fitting, pattern recognition, and prediction of data from NASA telescopes.
- Conducted multiple analyses to identify patterns in spectra and light-curves that allowed the quantification of magnetization of plasma in blazars.
- Implemented sophisticated numerical tools and data handling to C-SPEV that could calculate both discrete and continuous spectra from particle distributions with arbitrary shape, without impacting simulation runtime.
- Published 2 first-author papers.

Publications

Articles.....

- [12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.
- [11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.
- [10] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, *ApJ* **976**, 182, (2024), arXiv:2405.17581.
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, *MNRAS* **513**, 5766–5779, (2022), arXiv:2201.07790.
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, *Phys. Rev. D* **106**, 083015, (2022), arXiv:2112.09817
- [6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, *ApJ* **919**, 95, (2021), arXiv:2106.05356
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, *MNRAS* **501**, 4092–4102, (2021), arXiv:2009.02273.
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, *MNRAS* **496**, 974–986, (2020), arXiv:1910.14049.
- [3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, *MNRAS* **468**, 1169–1182, (2017), arXiv:1612.06383.
- [2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, *MNRAS* **438**, 1856–1869 (2014), arXiv:1310.5441.
- [1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, *Phys. Rev. D* **80**, 084023 (2009), arXiv:1009.1250.

Proceedings.....

- 5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, *Astron. Nachr.*, 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), arXiv:2011.13797.
- 4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, *Astron. Nachr.*, 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), arXiv:2011.13805.
- 3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, *PoS(SWIFT 10)* **233**, 159 (2014), arXiv:1502.07882.
- 2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, *EPJ Web Conf.* **61**, 02007 (2013), arXiv:1309.4612.
- 1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, *J. Phys.: Conf. Ser* **454**, 012001 (2013), arXiv:1211.1794.

Research Grants

NASA Fermi Cycle-12 Guest Investigator Program

A simple model to understand the blazar sequence, PI: Giannios, D., Co-I: **Rueda-Becerril, J. M.**

Grant #121077

2019

Grants and Fellowships

Oct. 2018 – Nov. 2020: Fellowship from the Mexican Federal Government for international postdoctoral studies awarded by the National Council of Science and Technology (CONACyT).

Jan. – Sep. 2018: Fellowship from the Mexican Federal Government under the *Program for the Professional Development of Higher Education Institutions*, awarded by the Secretariat of Public Education.

Sep. 2014 – Aug. 2016: Fellowship from the Mexican Federal Government to study abroad awarded by the National Council of Science and Technology (CONACyT).

Oct. 2011 – Jun. 2014: Fellowship *Santiago Grisolia* awarded by the Council of Education, Research, Culture and Sport of the Valencian Community, Spain.

Sep. 2009 – Aug. 2011: Fellowship for MSc studies at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo, granted by the Mexican Council of Science and Technology (CONACyT).

Jun. – Aug. 2007: Fellowship for a temporary stay (3 months) in a national research center under the *XVII summer of scientific investigation program* awarded by the Mexican Academia of Science.

Projects

Blazar Population ML

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Developing forecasting pipeline for blazar gamma-ray flares using a time series foundation model, fine-tuned with synthetic light curves generated with Tleco.

WindsOfChange

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++)

github.com/UW-ForestMycobiomeLab/WindSporesHills

- Developing C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrains with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library (Python/Rust)

github.com/zkdavis/Tleco

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONAHCYT Funded)

github.com/altjerue/paramo

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.
- Prototyped GPU acceleration using OpenACC directives on core Fokker-Planck solver kernels.

PatchworkMHD

Feb 2021 — Apr 2022

Large-Scale MHD Simulation Code (NSF Project)

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

Awards

Marcos Moshinsky Award: for *Best Poster* presented at the IWARA 2020 Video Conference, Mexico City, 6 – 12 September 2020.

Lic. Juan Josafat Pichardo Cruz Award: for finishing and defending a licentiate thesis within a year after completing the undergraduate credits, granted by the Universidad Autónoma del Estado de México, 2009.

Invited Talks

The Role of Machine Learning in Environmental Science

Seminar on Climate Change Ecology, University of Washington, Seattle, WA, USA, March, 2024

Morphology of the spectra from numerical simulations of the internal shocks model for blazars

Astrophysics Seminar, Purdue University, West Lafayette, IN, USA, February 4, 2019

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars

DATA group weakly Seminar, Instituto de Astronomía, UNAM, Mexico City, Mexico, June 19, 2018

Numerical treatment of non-thermal radiation in the internal shocks model for blazars

Weekly Seminar, Instituto de Física y Matemáticas, Morelia, Mexico, March 2, 2018

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars

IVICFA's Fridays: Computation in Physics, IFIC, Paterna, Spain, October 17, 2014

Meetings and conferences

Contributed Talks.....

Simulations of supermassive binary black holes accretion dynamics in the spinning case

APS April Meeting 2022, New York City, NY, April 9–12, 2022

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows

APS April Meeting 2021, April 17–20, 2021

The blazar sequence revised

9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020

<https://www.youtube.com/watch?v=BAZNWLNT69M>

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario

Extreme-Astrophysics in an Ever-Changing Universe, Ierápetra, Greece, June 16–20, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets

XXXIV Biennial meeting of the Royal Spanish Society of Physics, Valencia, Spain, July 15–19, 2013

Poster Sessions.....

A numerical approach for radiative cooling in relativistic outflows

9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020

Marcos Moshinsky Award for Best Poster. <https://www.youtube.com/watch?v=0TJiKg7k0PI>

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars

Swift: 10 years of Discovery, Rome, Italy, December 2–5, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets

The Innermost Regions of Relativistic Jets and Their Magnetic Fields, Granada, Spain, June 10–14, 2013

Teaching & Mentoring Experience

Zachary Davis [8, 9]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2022

Amanda O. Harrison [5]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2020

Hao Zhang [4]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2019

Thermodynamics (Graduate Level)

Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo

Substitute Instructor

Jun 2018

Professional Development

High Performance Computing on Frontera

Jason Allison et al., TACC, Austin, TX, USA, May 20, 27 and June 3, 2021

Lecture

Writing Winning Grants

Dr. Lauren Broyles, Purdue University, West Lafayette, IN, USA, November 7, 2019

Lecture

XSEDE HPC Workshop: Summer Boot Camp

John Urbanic, Purdue University, West Lafayette, IN, USA, June 3 – 6, 2019

Workshop

Data Analysis and Machine Learning with Python

Dr. Alejandro Torres, Universitat de València, Burjassot, Spain, February 7 – 16, 2017

Workshop

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit

Dr. Vassilios Mewes, Universitat de València, Burjassot, Spain, July 6 – 7, 2016

Workshop

No. of hours: 8

The Universe in the Light of PLANCK and BICEP2

Prof. Nick Mavromatos, Universitat de València, Burjassot, Spain, May 23 – 16, 2014

Lecture series

No. of credits: 2

Dark Matter

Prof. Alejandro Ibarra, Universitat de València, Burjassot, Spain, September 23 – 27, 2013

Lecture series

No. of credits: 2

International Cagès School on Cosmic Accelerators

Institut d'Études Scientifiques de Cargès, Cargès, France, April 23 – May 8, 2013

Summer school

Introduction to C++ Programming

Dr. Jacek Generowicz, Universitat de València, Burjassot, Spain, April 9 – 12, 2012

Workshop

No. of credits: 6

Numerical Relativistic Astrophysics

Prof. Luciano Rezzolla, Universitat de València, Burjassot, Spain, March 27 – April 4, 2012

Lecture series

No. of hours: 9

Fortran for Scientific Computing

HLRS, University of Stuttgart, Stuttgart, Germany, Mar. 5 – 9, 2012

Workshop

No. of hours: 33

Outreach

Los más rápidos y los más furiosos (*The Fastest and the Most Furious*)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco
Tabasco, Mexico

Online talk
September 4, 2020

Una simulación de la física y la astrofísica (*A Simulation of Physics and Astrophysics*)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco
Tabasco, Mexico

Online talk
August 14, 2020

ANITA y la teoría de los universos paralelos (*ANITA and the Theory of Parallel Universes*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/anita

Blog post
May 29, 2020

¿Qué es el modelo SIR? (*What is the SIR Model?*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/modelo-sir

Blog post
May 25, 2020

Evolución del brote epidémico de COVID-19 (*Evolution of the COVID-19 Epidemic Outbreak?*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/covid19
Collaborator with the data analysis/modeling

Blog post
April 5, 2020

Annual Department of Physics and Astronomy Poster Event

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Posters (3) presentation
November 13, 2019

Post-Doc Panel Q&A: What Happens When we Complete our PhDs?

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Panelist
April 10, 2019

Annual Department of Physics and Astronomy Poster Event

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Poster presentation
November 14, 2018

¿Decía Einstein la verdad? (*Was Einstein Telling the Truth?*)

Facultad de Ciencias, Universidad Autónoma del Estado de México
Toluca, Mexico

Talk
March 11, 2009

Synergetic Activities

X Scientific Meeting of the Spanish Astronomical Society

Organizing contributor, 14–16 December, 2012

Valencia, Spain

Other activities

Aug 2007– May 2009: Representative of the Physics students community at the Governing Council of the Faculty of Sciences of the Universidad Autónoma del Estado de México.

Languages

Spanish: Native proficiency

Catalan: Basic

English: Full professional proficiency

Nahuatl: Basic